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UNIT I – Introduction to OOP Concepts and Control Structure

1. What is a programming paradigm? Explain different types of programming paradigms.
2. Explain the difference between **procedural**, **object-oriented**, **functional**, and **logic/rule-based** paradigms.
3. Why do we need Object-Oriented Programming (OOP)?
4. Explain the main features of OOP:
   - Classes and Objects
   - Data abstraction and encapsulation
   - Inheritance
   - Polymorphism
5. What are the **benefits of OOP**?
6. Explain **Java** as an Object-Oriented Programming language.
7. Write the **structure of a simple Java program** and explain each section.
8. Explain the role of **JVM (Java Virtual Machine)** in Java program execution.
9. Differentiate between **primitive types** and **reference types** in Java.
10. Explain **operators and expressions** in Java with examples.
11. What is the **Java Class Library**?
12. Explain **control statements** in Java with syntax and examples:
    - if, if-else, nested if
    - for, while, do-while, for-each
    - break, continue
13. Case Study: Compare Java's OOP features with another OOP language (like C++ or Python).

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UNIT II – Introduction to Classes, Objects and Arrays

1. What is a **class** and what is an **object**?
2. Explain **field declaration** and **method declaration** with examples.
3. How do you **create and access** class members?
4. Explain how to **pass objects as parameters** and **return objects from methods**.
5. What are **set** and **get** methods? Give an example.
6. Explain **constructors** and the use of the **this keyword**.
7. What are **static methods and variables**?
8. Explain **method overloading** with example.
9. What is the **scope of declaration** in Java?
10. What is a **Java API package**? Give examples.
11. Explain **arrays in Java**: declaration, creation, and initialization.
12. Explain **multidimensional arrays** and **variable-length argument lists**.
13. How are **command-line arguments** used in Java?
14. Explain **input/output streams** in Java: Byte streams and Character streams.
15. What is the **PrintWriter class** used for?
16. Case Study: Explain how a **Banking System** or **Library Management System** can be implemented using classes, objects, and arrays.

UNIT III – Inheritance, Polymorphism, Exception Handling, and Multithreading

1. What is **inheritance**? Explain its types with examples.
2. What are **superclasses** and **subclasses**?

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3. What is the use of the **protected** keyword?
4. How are **constructors** used in subclasses?
5. What is **polymorphism**? Explain **compile-time** and **run-time** polymorphism with examples.
6. What are **abstract classes** and **abstract methods**?
7. What are **final methods** and **final classes**?
8. What is **dynamic binding**? Explain with example.
9. What is an **interface**? How is it different from a class?
10. What is **exception handling**? Explain the need for it.
11. Explain keywords: try, catch, throw, throws, finally.
12. Write a program to demonstrate **multiple catch** and **nested try** blocks.
13. What are **built-in exceptions** in Java?
14. Explain the concept of **multithreading**.
15. How to **create a thread** in Java (by extending Thread class or implementing Runnable)?
16. Explain the **Java Thread Model** and **Main Thread**.
17. Case Study: Explain how **Online Banking or Library System** can be developed using inheritance, exception handling, and multithreading.

UNIT IV – Graphics Primitives, Scan Conversion, Windowing, and Clipping

1. What are **graphics primitives**?
2. Define **pixel**, **resolution**, **aspect ratio**, and **frame buffer**.
3. Explain different **display devices** used in computer graphics.
4. Explain **applications of computer graphics**.

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5. Explain **DDA line drawing algorithm** with steps and example.
6. Explain **Bresenham's line drawing algorithm** with example.
7. Explain **DDA circle drawing algorithm**.
8. Explain **Bresenham's circle drawing algorithm**.
9. Explain **Midpoint circle drawing algorithm**.
10. What is a **polygon**? Explain types: convex, concave, and complex.
11. What is **inside test** in polygons?
12. Explain **flood fill** and **seed fill** algorithms.
13. Explain **scan line polygon fill** algorithm.
14. What is **windowing** and **clipping**?
15. Explain **Cohen–Sutherland line clipping algorithm**.
16. Explain **Sutherland–Hodgeman polygon clipping algorithm**.
17. Case Study:
    - Real-Time Map Rendering using line and circle drawing algorithms.
    - 3D pipeline and polygonal modeling in real-world applications.

UNIT V – 2D, 3D Transformations and Projections

1. What is **2D transformation**? Explain types:
    - Translation
    - Scaling
    - Rotation
    - Shear
2. Explain **rotation about an arbitrary point** in 2D.
3. What is a **3D transformation**? Explain:
    - Translation
    - Scaling
    - Rotation
    - Shear

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4. Explain **rotation about an arbitrary axis** in 3D.
5. What is a **projection**? Why is it used in computer graphics?
6. Differentiate between **parallel** and **perspective** projections.
7. Explain types of **parallel projections**:
    - Oblique: Cavalier and Cabinet
        - Orthographic:

        Isometric, Diametric,

        Trimetric
8. Explain **1-point, 2-point, and 3-point perspective projections**.
9. Compare **2D and 3D transformations**.